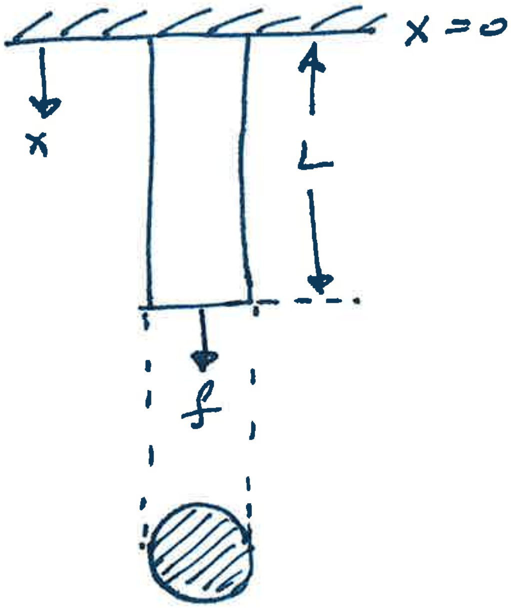
EXTENSION

Consider a ~~mass~~ weight-less rod



Axial displacement

$u_x = u(x) = 0$ @ $x = 0$

$u_x \neq 0$ @ $x = \frac{L}{2}$

$u_x = \max(u_x)$ @ $x = L$ → (1)

Recall LEHI behavior

$\epsilon_{xx} = \frac{\partial u_x}{\partial x}$, $\epsilon_{yy} = \frac{\partial u_y}{\partial y}$, $\epsilon_{zz} = \frac{\partial u_z}{\partial z}$

$\epsilon_{xy} = \epsilon_{yx} = \frac{1}{2}\left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right)$ $\epsilon_{yz} = \epsilon_{zy} = \frac{1}{2}\left(\frac{\partial u_x}{\partial z} + \frac{\partial u_x}{\partial x}\right)$

$\epsilon_{xz} = \epsilon_{zx} = \frac{1}{2}\left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x}\right)$

For 1D case: (Apply) $\boxed{\epsilon_{xx} = \frac{\partial u_x}{\partial x}}$ → (2)

Displacement: $\int_0^x \epsilon_{xx}\, dx = \int_0^x \frac{\partial u_x}{\partial x}\, dx$

$= u_x(x) - u_x(0)$

From (1): $u_x(0) = 0$

$\Rightarrow$ ~~uxcan~~ $\boxed{\int_0^x \epsilon_{xx}\, dx = u_x(x)}$ → (3)

From Hooke's LEHI behavior:

$\epsilon_{xx} = \frac{1}{E}[\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})]$, $\epsilon_{zz} = \frac{1}{E}[\sigma_{zz} - \nu(\sigma_{xx} + \sigma_{yy})]$,

$\epsilon_{yy} = \frac{1}{E}[\sigma_{yy} - \nu(\sigma_{xx} + \sigma_{zz})]$, $\epsilon_{xy} = \frac{1}{2G}\sigma_{xy}$, $\epsilon_{xz} = \frac{1}{2G}\sigma_{xz}$,

$\epsilon_{yz} = \frac{1}{2G}\sigma_{yz}$

1D case:

$$\epsilon_{xx} = \frac{1}{E}\left[\sigma_{xx} + \nu(\underbrace{\cancel{\sigma_{yy}}}_{0} + \underbrace{\cancel{\sigma_{zz}}}_{0})\right] \quad \text{where } \sigma_{yy} = \sigma_{zz} = 0$$

$$\Rightarrow \epsilon_{xx} = \frac{\sigma_{xx}}{E} \rightarrow (4) \quad \Rightarrow \epsilon_{xx} = \frac{f}{AE} \rightarrow (5)$$

Substituting in Eq (3): $\int_0^x \frac{\sigma_{xx}}{E} = u_x(x)$

$$\Rightarrow \boxed{u_x(x) = \int_0^x \frac{f}{AE}\,dx} \rightarrow (6)$$

General case: $f = f(x)$, $A = A(x)$, $E = E(x)$

$$\Rightarrow \boxed{u_x(x) = \int_0^x \frac{f(x)}{A(x)\,E(x)}\,dx} \rightarrow (7)$$

Special case: {f, A, E are independent of X
∴ we seek u_x @ $x = L$ i.e., deflection δ }

$$\Rightarrow u_x(L) = \delta = \frac{f}{AE}\int_0^L dx$$

$$\Rightarrow \boxed{u_x(x=L) = \delta = \frac{fL}{AE}} \rightarrow (8)$$

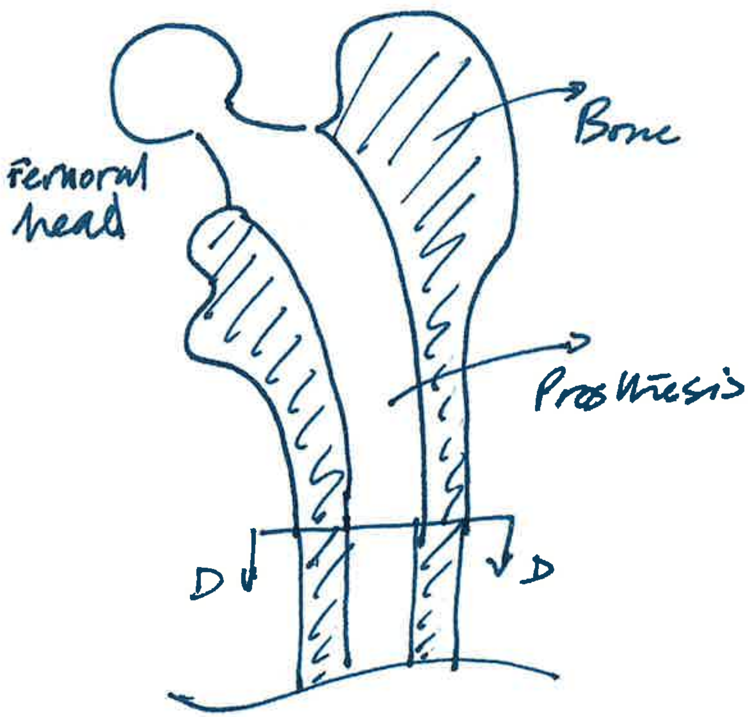
Other special cases:

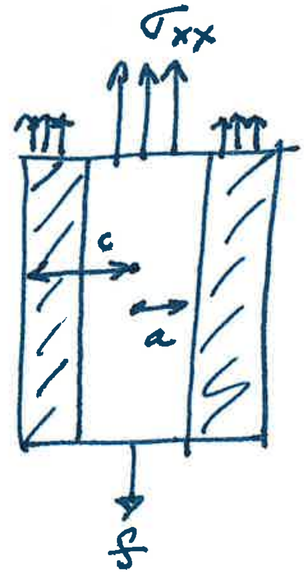
Homogenous Rod	$E \neq E(x)$	$u_x(x) = \frac{1}{E}\int_0^x \frac{f(x)}{A(x)}\,dx$
Constant crossection	$A \neq A(x)$	$u_x(x) = \frac{1}{A}\int_0^x \frac{f(x)}{E(x)}\,dx$
Constant load	$f \neq f(x)$	$u_x(x) = \frac{1}{f}\int_0^x \frac{1}{A(x)\,E(x)}\,dx$

General case: $u_x(x=c) - u_x(x=a) = \int_a^c \frac{f(x)}{A(x)\,E(x)}\,dx$

Integration: linear operator ∴ $\int_a^c \frac{f(x)}{A(x)\,E(x)}\,dx = \int_a^b \frac{f(x)}{A(x)\,E(x)}\,dx + \int_b^c \frac{f(x)}{A(x)\,E(x)}\,dx$

Prosthetic Hip Implants





Let part of load (f) carried by bone be f_b

& prosthetic be f_p

Axial Equilibrium $\Rightarrow f = f_p + f_B \rightarrow ①$

Mean axial stresses $\sigma_{xx}^{b} = \dfrac{f_b}{A_b} \rightsquigarrow ②$ $\quad \sigma_{xx}^{p} = \dfrac{f_p}{A_p} \rightsquigarrow ③$

Key question: What is f_b & f_c ?

From $f = f_p + f_b$ eq ① we have

→ one equation
→ two unknowns } statically indeterminate problem

We need a second equation — Assume
→ Assume no relative movement
→ for a painless implant

$\boxed{\epsilon_{xx}^{p} \equiv \epsilon_{xx}^{b}}$ ④

→ Uniform properties along length of prosthesis

For Prosthesis

$$u_x(x=L) - u_x(x=0)$$

$$= \int_0^L \epsilon_{xx}^{p}\, dx = \int_0^L \frac{\sigma_{xx}^{p}}{E_p}\, dx$$

$$= \int_0^L \frac{f_p}{A_p E_p}\, dx$$

With $u_x(x=0) = 0$ we have $u_x(x=L) = \delta_p$

$$\Rightarrow \delta_p = \int_0^L \frac{f_p}{A_p E_p}\, dx = \frac{f_p}{A_p E_p}\int_0^L dx$$

$$\Rightarrow \boxed{\delta_p = \frac{f_p L}{A_p E_p}} \rightarrow (5)$$

For Bone:

$$\boxed{\delta_b = \frac{f_b L}{A_B E_B}} \rightarrow (6)$$

To ensure compatible displacements:

$$\delta_p = \delta_b \rightarrow (7)$$

$$\Rightarrow \frac{f_p}{A_p E_p} = \frac{f_b}{A_b E_b}$$

$$\Rightarrow \boxed{f_p = \frac{A_p E_p}{A_b E_b} f_b} \rightarrow (8)$$

From equation (1): $f = f_p + f_b$

Substitute (8) in (1)

$$\Rightarrow f = \frac{A_p E_p}{A_b E_b} f_b + f_b = f_b\left(1 + \frac{A_p E_p}{A_b E_b}\right)$$

$$\Rightarrow \boxed{f_b = \frac{f A_b E_b}{A_b E_b + A_p E_p}} \rightarrow (9) \quad \text{ly} \quad \boxed{f_p = \frac{f A_p E_p}{A_b E_b + A_p E_p}} \rightarrow (10)$$

Substitute (9) & (10) in (2) & (3), respectively

$$\sigma_{xx}^{p} = \frac{1}{\cancel{A_p}}\left(\frac{\cancel{A_p} E_p f}{A_p E_p + A_b E_b}\right) = \frac{E_p f}{A_p E_p + A_p E_b} \rightarrow (11)$$

$$\sigma_{xx}^{b} = \frac{1}{\cancel{A_b}}\left(\frac{\cancel{A_b} E_b f}{A_p E_p + A_b E_b}\right) = \frac{E_b f}{A_p E_p + A_p E_b} \rightarrow (12)$$

Special case:

$E_p = E_b = E$ Young's moduli are matched!

$A_p + A_b = A$

$$\Rightarrow \sigma_{xx}^{p} = \frac{\cancel{E} f}{(A_p + A_b)\cancel{E}} = \frac{f}{A}$$

|||ly

$$\sigma_{xx}^{b} = \frac{f}{A}$$

We recover

$$\boxed{\sigma_{xx} = \frac{f}{A}}$$